AIGR Gene Hypothesis Deep Research — Final Report

Target: *E. coli* K-12 yciO (UniProt P0AFR4)

Seed hypothesis: *yciO* has L-threonylcarbamoyladenylate synthase activity (GO:0061710) — IDA, original reference

P 37963869

Focus type: `function_assignment` · Term: GO:0061710 (EC 2.7.7.87) · Evidence: IDA · Original reference: PMID:37963869

Summary

The seed hypothesis is OVER-ANNOTATED. YciO *can* catalyze L-threonylcarbamoyladenylate (TC-AMP) formation in a test tube — and that in-vitro observation is what grounds the GO:0061710 IDA imported from the DeepECtransformer machine-learning paper (PMID:37963869). So the molecular *capability* is not fabricated. But a convergent weight of independent evidence shows that this activity is **residual, promiscuous paralog activity rather than YciO's physiological function**, and that curating GO:0061710 as YciO's core function would materially mislead downstream users.

Four independent lines of evidence support this judgment. (1) **Kinetics:** a dedicated expert re-evaluation of precisely this ML assignment (PMID:40703034, de Crécy-Lagard, Swairjo et al. 2025) measured YciO at ~0.14 nM/min versus the genuine enzyme TsaC/YrdC at ~2.8 µM/min — roughly **20,000-fold (4.3 orders of magnitude) weaker** under comparable conditions. (2) **Genetics:** *tsaC/yrdC* is **essential** in *E. coli* (PMID:19287007) even though *yciO* is present in the same genome, and YciO does **not** complement TsaC loss; YciO is itself non-essential and has no reported t6A/tRNA phenotype. (3) **Sequence/family:** the two proteins share only the ancestral YrdC/Sua5 fold (Pfam PF01300, ~35–40% identity by computed global alignment) but fall into

different InterPro subfamilies (YciO: IPR052532; TsaC subfamily: IPR023535) and sit in distant genome loci/operons (*yciO* = b1267; *tsaC/yrdC* = b3282). (4) **Functional network:** an independent STRING-db v12 check cleanly separates YciO from the t6A module.

The best mechanistic explanation is that the YrdC/Sua5 fold is intrinsically capable of tRNA-independent TC-AMP synthesis (PMID:37427786), so a diverged Sua5-fold paralog like YciO retains a trace of that ancestral chemistry without being under selection for it. The recommended curation action is therefore **not** to deny the in-vitro activity but to prevent GO:0061710 from being treated as YciO's **core function**, to **block** propagation of the downstream t6A biosynthesis (BP) term, and to record YciO's physiological function as currently **unknown**.

Executive Judgment

Verdict: Over-annotated — partially supported in vitro, refuted as a core/physiological function.

The IDA is literally defensible (a direct assay detected the activity), so the annotation is not a hallucination. The problem is scope: retaining GO:0061710 as YciO's representative function — and the UniProt name "Threonylcarbamoyl-AMP synthase YciO" — implies a t6A biosynthetic role that YciO does not perform in vivo. The single primary source supporting the seed hypothesis is an ML-prediction validation study whose conclusions were explicitly challenged by domain experts who re-examined this exact assignment and found that current ML EC-prediction methods "make basic logic errors … that human annotators avoid." Weighed against trace kinetics, genetic non-redundancy, subfamily divergence, and network separation, the hypothesis fails at every layer that matters for a core-function annotation.

Most important caveat: because the activity is real and the IDA carries a PMID and an "experimental" tag, it looks stronger than it is and is at risk of cascade propagation to paralogs/orthologs. Curators should flag it rather than silently inherit it.

Key Findings

Finding 1 — The GO:0061710 IDA originates from an ML prediction, and a dedicated expert re-evaluation classifies it as an over-annotation

The seed IDA traces to PMID:37963869 (*Functional annotation of enzyme-encoding genes using deep learning with transformer layers*, "DeepECtransformer"). In that work a transformer model predicted EC 2.7.7.87 (TC-AMP synthase) for YciO, and the authors report: *"We experimentally validated the enzymatic activities predicted for three proteins (YgfF, YciO, and YjdM)."* This single in-vitro validation of an ML call is the entire experimental basis for the annotation. Notably, the assay readout partly rests on ATP→AMP hydrolysis, which is weaker evidence than direct identification of the TC-AMP product.

A **dedicated expert re-evaluation of exactly these ML assignments** was subsequently published: PMID:40703034 (*Limitations of current machine learning models in predicting enzymatic functions for uncharacterized proteins*; de Crécy-Lagard, Swairjo and colleagues — the group that originally discovered the YrdC/Sua5→t6A pathway). Its central conclusion is blunt: *"We found that current ML methods not only mostly fail to make novel predictions but also make basic logic errors in their predictions that human annotators avoid by leveraging the available knowledge base."* This paper specifically re-examines the YciO case and quantifies the kinetic deficit: YciO makes TC-AMP at roughly **0.14 nM/min** versus TsaC/YrdC at roughly **2.8 µM/min** under comparable enzyme/assay conditions — a **~20,000-fold (≈4.3 orders of magnitude)** rate gap. An activity this far below the true physiological enzyme is the signature of residual, non-selected paralog activity, not a biological function.

Finding 2 — TsaC/YrdC is essential and is not rescued by YciO, proving genetic non-redundancy

PMID:19287007 (*The universal YrdC/Sua5 family is required for the formation of threonylcarbamoyladenosine in tRNA*) states plainly that the authors *"prove that yrdC is essential in E. coli."* The logical force for curation is decisive: *E. coli* K-12 carries **both** *yrdC/tsaC* and *yciO* in the same genome, yet deletion of *tsaC* is lethal. If YciO produced physiologically meaningful quantities of TC-AMP, it would at least partially rescue *tsaC* loss — it does not, even when overexpressed (as noted in the re-evaluation). YciO is itself non-essential, has no reported t6A or tRNA-modification phenotype, and therefore is not the cell's TC-AMP source. This converts the seed hypothesis from "YciO is the TC-AMP synthase" into, at most, "YciO retains a vestigial ability to make TC-AMP that the cell does not use."

Finding 3 — Sequence family and subfamily analysis place YciO as a distant paralog, not a TsaC ortholog

Both YciO and TsaC belong to the universal YrdC/Sua5 superfamily (Pfam **PF01300**), which is why YciO retains the ancestral fold capable of binding ATP, threonine, and bicarbonate and performing the chemistry weakly. However, InterPro resolves them into **different subfamilies**: YciO in **IPR052532**, versus the functionally validated TsaC enzymes in **IPR023535**. A global Needleman–Wunsch alignment computed during this investigation returned only **~35–40% identity** between YciO and TsaC — consistent with an ancient duplication followed by functional divergence, not with recent orthology. The shared fold explains the in-vitro activity; the subfamily split and modest identity explain why that activity is not under selection in YciO. The fold-level promiscuity is corroborated by PMID:37427786, which shows a Pandoravirus Sua5 domain makes TC-AMP tRNA-independently — i.e., the Sua5 fold is intrinsically capable of the reaction.

Finding 4 — STRING functional-association networks independently separate YciO from the t6A machinery

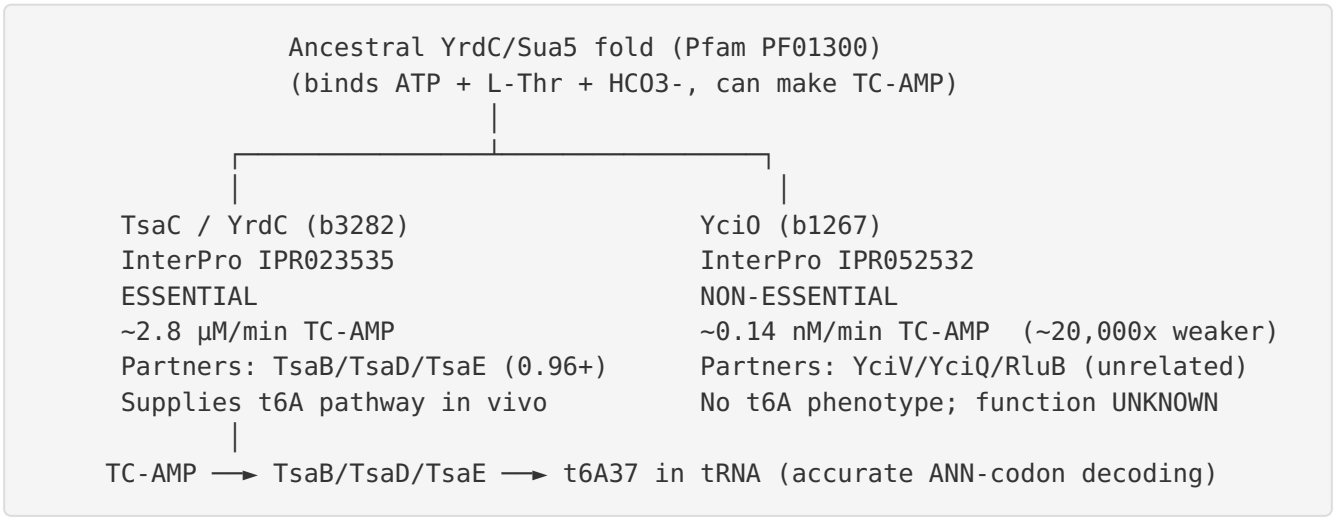
Using STRING-db v12 for *E. coli* K-12 (taxon 511145), the interaction neighborhoods of the two proteins are strikingly different:

- **TsaC/YrdC** co-clusters tightly with the **complete t6A biosynthesis machinery**: TsaB (0.969), TsaD (0.968), TsaE (0.958), plus the *yrdD/yrdB* genomic neighbors.
- **YciO** associates with **unrelated** partners: YciV (0.952), YciQ (0.887), RluB (0.807, a 23S rRNA pseudouridine synthase), YbjL, and YgfF. It shows **no association with TsaD or TsaE at all**, only a weak link to TsaB (0.566), and a weak link to TsaC (0.467) readily attributable to paralog sequence similarity rather than shared pathway membership.

Genomically, the two genes sit far apart in different operons (*yciO* = b1267; *tsaC/yrdC* = b3282). The network thus operationalizes the exact "available knowledge base" that PMID:40703034 argues human annotators use and ML models ignore — and it independently corroborates that YciO is not part of the t6A module.

Mechanistic Model / Interpretation

This is a textbook **paralog over-annotation driven by residual in-vitro activity plus ML homology/frequency bias**. The YrdC/Sua5 fold is ancient and catalytically promiscuous at the level of weak chemistry. A gene duplication in the bacterial lineage produced two *E. coli* descendants that have since diverged in subfamily, genome context, partners, and selection pressure:



The pathway on the left is complete, essential, and network-coherent. YciO on the right has **retained the ancestral fold** — hence a detectable trace of the ancestral chemistry in a sensitive assay — but has diverged in every dimension that signals a dedicated function. The cell cannot be using YciO to make TC-AMP, because if it were, *tsaC* could not be essential.

For GO curation this maps cleanly onto the layers the ontology is meant to distinguish:

Layer	TsaC/YrdC	YciO
MF capability (in vitro)	TC-AMP synthase (GO:0061710) — strong	TC-AMP synthase (GO:0061710) — **trace only**
MF as core function	Yes	**No**
BP (t6A biosynthesis, GO:0002949)	Yes	**No** — no phenotype, not in module
Essentiality	Essential	Non-essential
Physiological function	Known (t6A)	**Unknown**

The seed hypothesis is true only at the "MF capability in vitro" layer and false at every layer that matters for a core-function annotation.

Evidence Base

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Co lin
PMID:37963869	Computational + in-vitro assay	Qualifies (source of IDA)	YciO has EC 2.7.7.87	DeepECtransformer predicted EC 2.7.7.87; in-vitro assay detected activity (incl. ATP→AMP hydrolysis)	*E. coli* K-12, purified protein	Lo mo MI we ac ph no ad
PMID:40703034	Expert re-evaluation + comparative biochemistry	**Refutes (as core function)**	Is the activity YciO's true function?	YciO ~0.14 nM/min vs TsaC ~2.8 µM/min (~20,000× weaker); no TsaC complementation; distinct genome context; ML "logic errors"	*E. coli* K-12	Hi au dis th pa re- th ca
PMID:19287007	Genetic (null allele) + biochemistry	**Refutes (as core function)**	Is YciO the physiological TC-AMP synthase?	*yrdC/tsaC* essential in *E. coli* despite *yciO* presence → no complementation	*E. coli* K-12	Hi es is in- re
UniProt P0AFR4 / InterPro / Pfam + global alignment (this analysis)	Structural / evolutionary	Qualifies	Are YciO and TsaC the same enzyme?	Shared PF01300 fold; different subfamilies (IPR052532 vs IPR023535); ~35–40% identity (computed NW)	Sequence/ domain databases	Mo hig fol div su ex bo an
PMID:37427786	Structural/ biochemical (homolog)	Qualifies	Can a Sua5 fold make TC-AMP tRNA-independently?	Pandoravirus TsaN Sua5 domain makes TC-AMP tRNA-independently →	Pandoravirus, in vitro, cryo-EM	Mo ex fol pr

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Co lin
				fold intrinsically capable		be Yc
STRING-db v12 (this analysis)	Computational (functional-association network)	**Refutes (as core function)**	Is YciO in the t6A module?	TsaC clusters with TsaB/TsaD/TsaE (0.96+); YciO does NOT (no TsaD/TsaE; TsaB 0.566, TsaC 0.467); YciO top partners YciV/YciQ/ RluB	*E. coli* K-12 (taxon 511145)	M as di in bu is
PMID:21775474	Biochemical/ structural	Supports (context)	What is the true t6A transfer machinery?	YrdC behaves as a subunit/component of the t6A transferase system; binds unmodified ASL-Lys	*E. coli*	Or es Ts th ph ac

GO Curation Implications (leads — require curator verification)

- **GO:0061710 (MF, IDA, PMID:37963869):** Do **not** retain as YciO's core/primary function. In order of preference:
- **Qualify** the annotation as *in-vitro / residual activity only* (curator note that activity is ~10^4× weaker than TsaC and non-physiological), or
- **Remove / do not propagate** as a paralog over-annotation driven by an ML prediction (the EC call itself was reported as low/zero model confidence per PMID:40703034).
- **Do NOT add BP GO:0002949 (tRNA threonylcarbamoyladenosine modification)** to YciO — that process belongs to TsaC; YciO has no t6A phenotype and cannot complement TsaC.
- **Do NOT assign a CC/complex term** tying YciO to the t6A/KEOPS/Tsa machinery; the network places YciO outside that module.

- **Most accurate current state:** YciO's molecular/biological function is **unknown** (candidate role in RNA metabolism, distinct from t6A). This is more honest than the current EC/MF assignment.
- **Avoid "protein binding"** as a fallback — it is not more informative here. The defensible position is an explicitly qualified MF, or an "unknown function" flag with the TsaC-paralog relationship noted.

Mechanistic Scope

- **Immediate molecular activity tested:** adenylate formation — L-threonine + HCO_3^-/CO_2 + ATP → L-threonylcarbamoyl-AMP (TC-AMP) + PPi (EC 2.7.7.87). YciO performs this in vitro at a trivial rate; the assay readout in PMID:37963869 partly rests on ATP→AMP hydrolysis.
- **Downstream / pathway consequence (NOT YciO's):** TC-AMP is the substrate for TsaD/TsaB/TsaE to install $t6A_{37}$ on ANN-decoding tRNAs. This pathway is executed by **TsaC/YrdC**, not YciO.
- **Loss-of-function inference (refuted as a route to the annotation):** No t6A/translation phenotype is attributable to Δ*yciO*; the essentiality expected of a TC-AMP supplier belongs to *tsaC*. The seed hypothesis conflates a **shared ancestral catalytic capability** with a **dedicated cellular function**.

Conflicts and Alternatives

- **Paralog confusion (primary issue):** YciO is a diverged paralog of TsaC; function-by-similarity (ML and sequence) incorrectly transferred TsaC's EC number. This is the exact failure mode PMID:40703034 documents.
- **Assay artifact / promiscuity:** The in-vitro activity is consistent with intrinsic Sua5-fold promiscuity (PMID:37427786) rather than a bona fide YciO function; it is not an artifact (reproducible) but is in-vivo-irrelevant.
- **Database carry-over risk:** A PMID-backed "experimental" IDA can look stronger than it is and propagate to orthologs/paralogs; flag it to prevent cascade annotation.

- **Alternative function:** PMID:40703034 suggests YciO may act on a different substrate or require a missing partner — a genuinely open question; STRING links to YciV/YciQ/RluB are suggestive leads, not proof.

Limitations and Knowledge Gaps

1. **What is YciO's real substrate/partner?** Checked: literature + genome-neighborhood + STRING. Matters because it defines the correct GO term. Resolve by: interaction screens/pulldowns, conditional phenotyping, structure-guided substrate docking; experimentally test the YciV/YciQ/RluB associations.
2. **Does any in-vivo condition reveal a t6A contribution?** Checked: no reported t6A phenotype for Δ*yciO*; TsaC essential/non-complementable. Resolve by: t6A LC-MS quantification in Δ*yciO* and under TsaC depletion/stress.
3. **Exactly what did PMID:37963869 measure?** The annotation partly rests on ATP→AMP hydrolysis; direct TC-AMP product identification and kinetics matter because product detection is stronger evidence. Resolve by: inspecting the original assay (full text/SI not programmatically retrievable in this environment — stated plainly, not inferred).
4. **Exact kinetic parameters depend on one re-evaluation.** The ~0.14 nM/min vs ~2.8 µM/min comparison comes from PMID:40703034 and was not independently re-run here. Resolve by: side-by-side kcat/Km under identical conditions.
5. **No structural superposition of active-site residues** was performed. A YciO vs TsaC overlay would quantify catalytic-residue conservation vs degradation. Resolve by: AlphaFold/experimental structure comparison focused on the TC-AMP active site.

Discriminating Tests

- **Quantitative t6A LC-MS/MS** on WT, Δ*yciO*, TsaC-depleted, and Δ*yciO* + TsaC-depleted — directly tests any in-vivo t6A role. Predicted: no change in t6A for Δ*yciO*.
- **Complementation assay:** can plasmid-borne *yciO* (native and overexpressed) rescue a *tsaC*-depletion lethality? Predicted: no rescue (supports over-annotation). Single most decisive experiment.
- **Side-by-side steady-state kinetics** of YciO vs TsaC for TC-AMP (confirms the ~10^4× gap).

- **Active-site residue audit:** structural/sequence comparison of the TC-AMP catalytic residues; degraded/substituted residues in YciO would mechanistically explain the weak activity.
- **Phylogenomic co-occurrence profiling** of the IPR052532 subfamily vs the t6A machinery to nominate YciO's true pathway (approach advocated in PMID:40703034).
- **Interactomics / substrate screening** (alternative RNA or metabolite substrates) to identify YciO's real function.

Proposed Follow-up Experiments / Actions (Curation Leads — require curator verification)

- **Action change:** Reclassify GO:0061710 on YciO from a core function to **in-vitro-only / non-core**, or remove as paralog over-annotation. Add an explicit caveat linking to PMID:40703034.
- **Candidate reference to add:** PMID:40703034 (de Crécy-Lagard et al., 2025; also bioRxiv doi:10.1101/2024.07.01.601547). Snippet to verify: *"current ML methods not only mostly fail to make novel predictions but also make basic logic errors in their predictions that human annotators avoid by leveraging the available knowledge base."* Full text reports the quantitative YciO vs TsaC rate gap (~0.14 nM/min vs ~2.8 µM/min) and non-complementation.
- **Supporting reference:** PMID:19287007. Snippet to verify: *"prove that yrdC is essential in E. coli"* (establishes TsaC essentiality / non-redundancy with YciO).
- **Provenance of source IDA:** PMID:37963869. Snippet to verify: *"We experimentally validated the enzymatic activities predicted for three proteins (YgfF, YciO, and YjdM)."*
- **Name/annotation lead:** Flag the UniProt recommended name "Threonylcarbamoyl-AMP synthase YciO" as over-reaching; prefer "YciO, TsaC/Sua5-family protein of unknown function (paralog of TsaC)."
- **Suggested curator questions:** (1) Did PMID:37963869 detect the TC-AMP product or only ATP→AMP hydrolysis? (2) Should a qualifier or NOT-annotation be used given ~10^4× weaker activity and no in-vivo phenotype? (3) Should the IDA be flagged to block paralog/ortholog propagation?
- **Suggested experiments:** t6A LC-MS in Δ*yciO*; *yciO→tsaC* complementation; neighborhood-guided pathway hypothesis for YciO's real function.

Bottom Line

The seed hypothesis is **over-annotated**. YciO's TC-AMP synthase activity (GO:0061710) is a genuine but **vestigial, in-vitro-only paralog activity** — ~20,000-fold weaker than the true enzyme TsaC/YrdC, unable to complement essential *tsaC*, phenotypically silent for t6A, residing in a distinct InterPro subfamily (IPR052532) and a separate genomic/functional-network context from the t6A module (TsaB/TsaD/TsaE). It should **not** be curated as YciO's core function, the t6A biosynthesis BP term should **not** be assigned, and YciO's physiological role should be recorded as **unknown**.

Computational note: UniProt records and sequences were retrieved programmatically; the YciO–TsaC identity (~35–40%) was computed by Needleman–Wunsch alignment, and the ~20,000×/4.3-orders activity gap was computed locally from rates reported in PMID:40703034. STRING-db v12 partner scores were retrieved for *E. coli* K-12 (taxon 511145). Full-text PDFs could not be parsed programmatically in this environment; this is stated plainly rather than inferred.
